23MID0018

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In [49]:
```
#a
#b
#c

import csv
import os
header = ["Name","CGPA","Certification_count","Internship_Exp_yrs"]
data = [
    ["Raja","9.11","3","3 month"],
    ["Murugan","9.20","2","2 yrs"],
    ["arun","8.20","6","5 yrs"],
    ["ram","7.20","1","2 yrs"]
]
with open("student.csv","w",newline ="") as file:
    writer = csv.writer(file)
    writer.writerow(header)
    writer.writerows(data)
path = os.path.abspath("student.csv")
print(f"Path of student.csv is {path}")
```

```
Path of student.csv is /home/matlab/student.csv
```

In [50]:
```
#d
with open("student.csv","w",newline ="") as file:
    writer = csv.writer(file)
    writer.writerow(["Nideesh","9.0","6","4yrs"])
    writer.writerow(["kamali","9.7","4","3yrs"])
    writer.writerow(["santhosh","8.0","2","1yrs"])
```

In [87]:
```
#e
import csv
```

```
dict_1 = {
    "Name " : "Rajavel",
    "CGPA ": 9.11,
    "Certification_count ":3,
    "Intern_exp" : "3 yrs"
}
field = ["Name","CGPA","Certification_count","Intern_exp"]
with open("students_dict.csv","w") as file:
    writer = csv.DictWriter(file,fieldnames=field)
    writer.writerow(dict)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[87], line 12
     10 with open("students_dict.csv","w") as file:
     11     writer = csv.DictWriter(file,fieldnames=field)
---> 12     writer.writerow(dict)

File /opt/anaconda/lib/python3.13/csv.py:226, in DictWriter.writerow(self, rowdict)
    225 def writerow(self, rowdict):
--> 226     return self.writer.writerow(self._dict_to_list(rowdict))

File /opt/anaconda/lib/python3.13/csv.py:221, in DictWriter._dict_to_list(self, rowdict)
    219     wrong_fields = rowdict.keys() - self.fieldnames
    220     if wrong_fields:
--> 221         raise ValueError("dict contains fields not in fieldnames: "
    222                          + ", ".join([repr(x) for x in wrong_fields]))
    223 return (rowdict.get(key, self.restval) for key in self.fieldnames)

ValueError: dict contains fields not in fieldnames: 'CGPA ', 'Name ', 'Certification_count '
```

In [52]:
```
#f
header = ["Name","CGPA","Certification_count","Internship_Exp_yrs"]
with open("student_header.csv","w",newline ="") as file:
    writer = csv.writer(file)
    writer.writerow(header)
    writer.writerows(data)
path = os.path.abspath("student_header.csv")
print(f"Path of student.csv is {path}")
```

```
Path of student.csv is /home/matlab/student_header.csv
```

In [53]:
```
#g
new_rec =  [
    ["sam","8.11","7","3 month"],
["sachin","6.6","6","3 yrs"],
    ["ram","7.7","5","2 yrs"]
]
with open("student.csv","a") as file:
    writer = csv.writer(file)
    writer.writerows(new_rec)
path = os.path.abspath("student.csv")
print(f"Path of student.csv is {path}")
```

```
Path of student.csv is /home/matlab/student.csv
```

In [54]:
```
#h
with open("student.csv","r") as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

```
['Nideesh', '9.0', '6', '4yrs']
['kamali', '9.7', '4', '3yrs']
['santhosh', '8.0', '2', '1yrs']
['sam', '8.11', '7', '3 month']
['sachin', '6.6', '6', '3 yrs']
['ram', '7.7', '5', '2 yrs']
```

In [55]:
```
# 2 . Using Pandas
# a. Load student details to a Dataframe
# b. Use to_csv() to write the the contents from a data frame to csv file
# c. Read from this csv print and print all contents using read_csv()
# d. Use the concatenation method to append few more rows to the csv file
# e. Read the updated file
```

In [73]:
```
#a
import pandas as pd
data ={
    "name" :["raja","murugan"],
    "CGPA" :["9.11","9.10"],
    "certificates" : [3,5],
```

```
    "Internship_exp" :[8,7]
}
df = pd.DataFrame(data)
```

In [74]:
```
#b
df.to_csv("student_details.csv",index=False)
```

In [76]:
```
#c
df = pd.read_csv("student_details.csv")
print(df)
```

```
      name  CGPA  certificates  Internship_exp
0     raja  9.11             3               8
1  murugan  9.10             5               7
```

In [114…
```
header = ["Name","CGPA","Certification_count","Internship_Exp_yrs"]
rec1 = pd.DataFrame([["ram","7.0",6,4]],columns=header)
pd.concat([df,rec1])
```

Out[114…

	name	CGPA	certificates	Internship_exp	Name	Certification_count	Internship_Exp_yrs
0	raja	9.11	3.0	8.0	NaN	NaN	NaN
1	murugan	9.1	5.0	7.0	NaN	NaN	NaN
0	NaN	7.0	NaN	NaN	ram	6.0	4.0

In [85]:
```
# e read updated file
df = pd.read_csv("student_details.csv")
print(df)
```

```
      name  CGPA  certificates  Internship_exp
0     raja  9.11             3               8
1  murugan  9.10             5               7
```

In [86]:
```
# 3. Using json
# a. Create a dictionary of student details as in Question 1
# b. Save the dictionary as a JSON file [Hint: use json.dump()]
# c. Iterate through the JSON file and print the contents
# d. Read a dictionary as a string and convert that into a Python Dictionary object using json.load()
# e. Print the contents of the dictionary
```

In [134…
```
#a
#b
import json
dict_1 = {
    "Name " : "Rajavel",
    "CGPA ": 9.11,
    "Certification_count ":3,
    "Intern_exp" : "3 yrs"
},

with open("student.json","w") as file:
    writer = json.dump(dict_1,file)
```

In [135…
```
# c
with open("student.json","r") as file:
    reader = json.load(file)
    for row in reader:
        print(row)
```

```
{'Name ': 'Rajavel', 'CGPA ': 9.11, 'Certification_count ': 3, 'Intern_exp': '3 yrs'}
```

In [138…
```
# d
rec1 = [{
    "Name " : "Rajavel",
    "CGPA ": "9.11",
    "Certification_count ":"3",
    "Intern_exp" : "3 yrs"
},{
    "Name " : "murugan",
    "CGPA ": "7.11",
    "Certification_count ":"4",
    "Intern_exp" : "4 yrs"
},
```

```
        ]
with open("student.json","w") as file:
    writer = json.dump(rec1,file)
with open("student.json","r") as file:
    reader = json.load(file)
```

In [139…
```
#e
for row in reader:
    print(row)
```

```
{'Name ': 'Rajavel', 'CGPA ': '9.11', 'Certification_count ': '3', 'Intern_exp': '3 yrs'}
{'Name ': 'murugan', 'CGPA ': '7.11', 'Certification_count ': '4', 'Intern_exp': '4 yrs'}
```

In [140…
```
# 4 Using Pandas
# a. Load a student details dictionary to a dataframe
# b. Use to_json() to save as a json file [hint: use orient]
# c. Read from the json file and print all the contents
# d. Inspect the data using head(), info(), describe() methods
# e. Access the second column and print its contents
# f. Apply filter on a column and print its contents
# g. Add a new column and load the values
# h. Demonstrate updating a value
# i. Write the updated JSON (dataframe) file
```

In [141…
```
#a
import pandas as pd
data ={
    "name" :["raja","murugan"],
    "CGPA" :["9.11","9.10"],
    "certificates" : [3,5],
    "Internship_exp" :[8,7]
}
df = pd.DataFrame(data)
```

In [142…
```
#b
df.to_json("student_details.json",index=False)
```

In [143…
```
# c
df = pd.read_json("student_details.json")
print(df)
```

```
     name  CGPA  certificates  Internship_exp
0    raja  9.11             3               8
1  murugan  9.10             5               7
```

In [144…
```
# d
df.head()
```

Out[144…

	name	CGPA	certificates	Internship_exp
0	raja	9.11	3	8
1	murugan	9.10	5	7

In [145…
```
# d
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 2 entries, 0 to 1
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   name            2 non-null      str    
 1   CGPA            2 non-null      float64
 2   certificates    2 non-null      int64  
 3   Internship_exp  2 non-null      int64  
dtypes: float64(1), int64(2), str(1)
memory usage: 91.0 bytes
```

In [146…
```
#d
df.describe()
```

Out[146…

	CGPA	certificates	Internship_exp
count	2.000000	2.000000	2.000000
mean	9.105000	4.000000	7.500000
std	0.007071	1.414214	0.707107
min	9.100000	3.000000	7.000000
25%	9.102500	3.500000	7.250000
50%	9.105000	4.000000	7.500000
75%	9.107500	4.500000	7.750000
max	9.110000	5.000000	8.000000

In [149…
```
#e
print(df["CGPA"])
```

```
0    9.11
1    9.10
Name: CGPA, dtype: float64
```

In [150…
```
#f
print(df["certificates"]>4)
```

```
0    False
1     True
Name: certificates, dtype: bool
```

In [155…
```
# G
df["remarks"] = df["certificates"]
```

In [152…
```
# h and i updated json
print(df)
```

```
      name  CGPA  certificates  Internship_exp
0     raja  9.11             3               8
1  murugan  9.10             5               7
```

In [ ]: